qm-notes

kayla

Week 2

Definitions

and

The probability of A and B happening, also the joint probability of $A \cap B$, is computed using the following general rule:

$$\Pr(A \cap B) = \Pr(A|B) \times \Pr(B) = \Pr(B|A) \times \Pr(A)$$

or

The probability of A or B happening (which includes all of *only* A, *only* B, and *both* A *and* B) is computed using the following general rule:

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

where we subtract any 'overlap' between A and B which was double counted in adding both.

Independence

An event A is *independent* of another event B iff the probability of A happening is not conditional on the probability of B happening. Formally, the conditional probability of A on B is equal to the marginal probability of A, and therefore equal to the probability of A on not-B. That is,

$$\Pr(A|B) = \Pr(A| \sim B) = \Pr(A)$$

When this happens, the joint probability of A and B is

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B)$$

Mutual exclusivity

An event C is *mutually exclusive* with another event D iff neither event can occur simultaneously. Formally,

$$\Pr(A \cap B) = 0 \therefore \Pr(A \cup B) = \Pr(A) + \Pr(B)$$

This also means that the conditional probabilities of both events on each other $\Pr(A|B) = \Pr(B|A) = 0$.

Collective exhaustivity

A set of events, $E_i = \{E_1, E_2, ..., E_n\}$, is collectively exhaustive when they add up to 1. Formally,

$$\Pr(\bigcup_{i=1}^{n} E_i) = \sum_{i=1}^{n} \Pr(E_i) = 1$$

Classical probability

Classical probability is the study of probability in the absence of data. Therefore, definitions and concept in this paradigm are built upon relations between discrete events and cardinalities of sets. In the absence of data, we conceptualise probability as the ratio between the *cardinality of the set of outcomes in an event* and that of *the sample space*, where the sample space is the set of all possible outcomes.

Letting $E = \{e_1, e_2, ..., e_m\}$ be some event with m outcomes and $S = \{e_1, ..., e_m, e_{m+1}, ..., e_n\}$ represent the sample space,

$$\Pr(E) = \frac{|E|}{|S|}$$

An illustrative example is the theoretical 'concept' of rolling an equal-probability die, let's say a cubical die. In practice, the running *empirical* probability of rolling a die will almost never be perfectly $\frac{1}{6}$ for each side. However—especially relevant given the presumption of equal probability—we can theoretically expect each outcome to have a probability of $\frac{1}{6}$, i.e., the ratio between the cardinalities of the *single-outcome* event* $E \in S$ (1) and the sample space $S = \{1, 2, 3, 4, 5, 6\}$ (6).

Simple and compound events

Simple events—like the *single-outcome event referenced above—are, well, only 'made up' of a single outcome. Formally, simple events are events whose set of outcomes is a singleton ($|E| = 1$). Outcomes themselves can be seen as events if we're interested in examining the outcomes in and of themselves; the outcome–event distinction would only be a conceptual one in such a case.

Compound events, on the other hand, can be made up of multiple outcomes. To return to the die roll example, a compound event would look something like 'rolling an odd number' or 'rolling a number smaller than three'.

From specific to general

Variables in political science are usually stochastic, where whether an outcome (an event happening/taking on a certain value) is true is 'up to chance' and its associated causal mechanisms *change the probability* of it happening.

From a frequentist perspective at least, causal-statistical accounts of political phenomena can thus be viewed as 'repeated trials' on the population, where the probability of an effect (some outcome) manifests as the number of units/observations for which this is true. Observing a causal phenomenon increasing/decreasing this number, then, can be seen as some cause resulting in a higher probability of effect being true.

Random variables

Let Y be some random variable on a set of outcomes O. As Y is a function,

$$Y : O \mapsto \{y_1, y_2, ...\}$$

where $\{y_1, y_2, ...\}$ is the **distribution** (also sample space). Within this distribution, n_1 and n_2 are **realisations** of the random variable. In addition, the **support** of this distribution is the subset of the distribution s.t. $\forall y \in$ support, $\Pr(y) > 0$. Lastly, the function which maps all the members of the distribution to their associated probabilities,

$$f : \{y_1, y_2, ...\} \mapsto \{\Pr(y_1), \Pr(y_2), ...\}$$

is the **distribution *function***.

$\Pr(A) = \sum_{i=1}^{n} \Pr(A|B_i) \times \Pr(B_i)$

Discrete probability distributions

Bernoulli distribution

Bernoulli distributions exclusively concern *binary variables*, describing the frequency of both potential outcomes over repeated observations. It can be represented as follows:

$$\Pr(Y = y|p) = \begin{cases} (1-p) & \text{for} \quad y = 0, \\ p & \text{for} \quad y = 1 \end{cases}$$

In the general form,

$$\Pr(Y = y|p) = p^y(1-p)^{1-y}$$

The Bernoulli distribution is based on the assumption that the trials are independent, which then needs to be justified/self-evident when applying it in analysis.

Binomial distribution

The binomial distribution broadens the size (number of trials) of the Bernoulli distribution$\Pr(Y = y|n, p) = \binom{n}{y} p^y(1-p)^{1-y} = \frac{n!}{y!(n-y)!} p^y(1-p)^{1-y}$

Poisson distribution

$\Pr(Y = y|\mu) = \frac{\mu^y}{y! \times e^{\mu}}$, where $\mu > 0$ is the expected value of the number of events and the variance $\sigma^2 = \mu$.

This PMF describe the number of times one observes 0 events, 1 event, etc. throughout a fixed period of time. Events that can be described by a Poisson distribution are

- discrete (integer count)
- independent (just like Bernoulli), and
- distributed with a *known mean.*

Negative binomial distribution

$\Pr(Y = y|r, p) = \binom{y+r-1}{y} p^y(1-p)^r$, where y is the number of successes (events), r is the number of failures (non-events), $y + r$ is the number of opportunities (or Bernoulli trials), and p is the probability of a success. Events that can be described by a negative binomial distribution

- are independent
- have two possible outcomes (binary)
- keep a constant probability of successes from one trial to the next
- continue until a total of r successes (/failures) are observed, where r is fixed

Negative binomial distributions allow applications to decouple the variance from the mean.

Continuous probability distributions

Normal distribution

$$\phi(x) \equiv f(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

Power-transformed distribution

$$\phi(x) \equiv f(x; \mu, \sigma^2, \lambda) = \frac{\lambda x^{\lambda-1}}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x^\lambda-\mu)^2}$$

When $\lambda < 1$, the right side of the graph is longer than than the left. When $\lambda > 1$, the left side is longer than the right. An asymmetry where one function of the graph is longer than the other is called skewness.

An example of a power-transformed distribution is the log distribution,

$$f(x; \mu, \sigma^2) = \frac{1}{x\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(\ln(x)-\mu)^2}$$

more distributions - logistic - duration (exponential, pareto, gamma, weibull)

Week 3

Definitions

- Estimand: the quantity of interest we seek to know about some *population*
- Parameter: the formalised form of an estimand, usually taking a fixed value
- Estimator: a function that takes a sample as input and outputs an estimate
- Estimate: the parameter of a sample as an approximation of an estimand
- Sampling variability: the 'deviation' of sampling means relative to one another
  - Implication: If it isn't zero, then there is (non-zero) sampling variation
- Sampling error: the deviation of a sampling mean relative to the (true) population mean
  - Implication: If it isn't zero, then the sample's point estimates are biased

On the enterprise of sampling

Sampling is fundamentally about trying to square a circle, with the square being the endeavour to seek out some quantifiable truth about a population and the circle being constraints to how much of the population we can feasibly reach out to. It means that we're fundamentally aspiring towards an impossibility, although it's still worth doing because the imperfect results ('estimates') we get out of it are still *useful* (a common theme in statistic analysis!)

Especially with things like voting, there is a fundamental/stochastic uncertainty because the variables of interest are 'random', not just between subjects but also within subjects across e.g., time. Therefore, even if we could 'sample' everyone (at which point it is no longer a sample), we cannot be uncertain about our estimate given the looming possibility of temporal changes. Besides, taking a snapshot of *everyone at the same time* is categorically not possible.

But of course we are usually only able to sample a small proportion of the population. That introduces a systematic component of uncertainty to our estimates: any sample is both 'different' from other samples and different (i.e., deviant) from the population.

- Asymptotic
- Finite-samples

Estimands, estimates, estimators...

A random sample, $\{Y_1, Y_2, ..., Y_n\}$, is a set of random 'variables'. It is a proper subset of the population where each member of the population has an equal probability of being selected. When you calculate a sample statistic, the statistical *function* is applied on the observed sample (the input). This function is called an *estimator* and its output an *estimate*. Formally, an estimator is a function

$$f : \{Y_1, Y_2, ..., Y_n\} \mapsto \mathbb{R}$$

where the estimates are a subset of the set of real values, $\mathbb{R}$.

In a (hypothetical) indefinite repeated-samples scenario, there will be variation in the mean on account of not every sample containing the exact same members Y_i. An estimate from a sample will not be identical to the population value, with the difference being the *sampling error*. Graphing these samples along the values it takes also gives us an indicator of the *sampling variability*.

To formalise our expression of uncertainty about our estimates, we use the standard error. Using prior knowledge of sampling variability on repeated samples as a starting point, we can expect that our different estimates across different repeated samples yields a probability distribution of all the different estimates, known as the *sampling distribution*. Calculating the spread of this distribution then gives us the *standard error*.

Important: Note that our usage of the term 'variable' here differs from that which is found in lay statistical usage. While Y is some random variable across all members of the population, Y_i is the function localised to each observation i. Thus, the term 'variable' here is more analogous to an observation.

Confidence intervals

A confidence interval can be interpreted as the range in which repeated samples, over the long run, will contain the true mean with some given probability (often 95%).

Important: A 95% confidence interval does not mean that there is a 95% probability for the population parameter to lie inside some *specific* confidence interval, $[y - m, y + m]$. Rather, it pertains to the *a priori generalised CI function* $[Y - 1.96\sqrt{n}, Y + 1.96\sqrt{n}] \quad \forall n$ prior to the value

y being specified. That is, before the sample is drawn, we have 95% confidence that the sample we draw contains the population mean μ. Once the value has been decided and the samples taken, the CI either contains the (unknown) population parameter or it doesn't, with no way to tell for sure - though of course, we can be less or more confident about it.

It is also possible to reframe this 'prospecting' confidence in retrospect: if we repeat the procedure indefinite times, the CIs we come up with, regardless of the exact sample estimates, will contain the sample mean 95% (or more generally, $100(1-\alpha)\%$) of the time. The more samples we take, the more convergent to 95% it gets.

However, this estimation has another pitfall: it assumes that we know the population's standard deviation σ. Oftentimes, we cannot, and what we have is instead the estimate $\bar{s}$. Since we cannot use this, we have to relax our assumption of normality and use the t distribution instead.

Finite sample properties of estimators

Here, we talk about biasedness, variance, and relative efficiency. Both of these pertain to an increasingly large number of *draws/trials*.

Unbiased

An estimator is unbiased when the mean of the estimator distribution, given some fixed sample size, converges to the expected value of the parameter over the course of an indefinite amount of draws. Formally,

$$E[W(\theta)] - W(\theta) = 0$$

Relative efficiency

(Relative) efficiency is when, given two unbiased estimators W_1 and W_2, the more efficient estimator W_1 has

- Variances smaller than/equal to the other's variances on all values of the parameter
- A strictly smaller variance on at least one value of tne parameter

Formally

$$\forall\theta, \mathrm{Var}(W_1) \leq \mathrm{Var}(W_2) \text{ and } \exists\theta : \mathrm{Var}(W_1) < \mathrm{Var}(W_2)$$

Asymptotic properties of estimators

Asymptotic measures like consistency pertain to an increasingly large *sample size*.

Consistency

Consistency is the property of an estimator producing increasingly accurate estimates as the sample size increases. That is,

$$\lim_{n\to\infty} E[W(\theta)] = W(\theta)$$

Hypothesis testing

We will never know for sure the (*a posteriori*) odds of some sample statistic/estimator at hand producing an erroneous sample estimate. However, we can find out the *a priori* probability that our (interval) estimate may not contain the estimand (population parameter).

Levels α of a statistical significance tests are simply some given probability thresholds of a Type I error occurring. Formally, α is the the null hypothesis being rejected given that it actually happens:

$$\alpha = \Pr(\text{reject } \ H_0 | H_0)$$

Extra notes on hypothesis testing

At the end of the day, probability is (a formalisation of) expectation and uncertainty. When we're 95% confident of 'some result', we implicitly mean that we 'expect it to be a lot more likely' and assign only 5% to the possibility that it doesn't pan out. In frequentist statistics, this translates to an extension of the coin flip paradigm: if the phenomenon behind this 'some result' were to be repeated infinite times, we would see it occurring 95% of the time and not occurring 5% of the time.

In hypothesis testing, confidence (1 - significance) levels indicate the likelihood that there is indeed a 'result' under the hood (pertaining to the population) when our sample gives us a result. We are therefore $x\%$ (oftentimes 95%) confident about a (*retrospective*,) conditional event. That is, we're $x\%$ confident of some underlying process being present in the population given that we observe it in a sample. That means that its complement, the $(1-x)\%$, refers to the underlying process *not* being present given that we observe it (which we already did, so 'not observing some phenomenon in the sample' is irrelevant).

The probability of the phenomenon being true for a population given that we observed the phenomenon in a sample is equivalent to the probability of H_0 being false given that H_0 is rejected, $\Pr(\neg H_0 | \text{reject } \ H_0)$. Conversely, the complement is equivalent to the probability of H_0 being *true* given that H_0 is rejected, $\Pr(H_0 | \text{reject } \ H_0)$.

Approaches to assessing uncertainty

Analytic method

This approach presumes a normal distribution and calculates confidence intervals in 'closed form', i.e., exclusively by manipulating known parameters mathematically. Statistical software calculates analytic estimates using the t distribution by default. This means that the analytic method works even for smaller samples, *provided that the normality assumption still holds.*

Simulation

This approach also presumes a normal distribution. However, unlike the analytic method, it generates multiple 'new' samples from scratch (i.e., it takes draws from the distribution) using the distribution parameters of existing sample data.

This approach makes tangible the frequentist idea of 'finite samples' while keeping computational power requirements low.

Importantly, this process only 'preserves' data if we can argue that the underlying DGP yields a normally distributed random variable. When that assumption is violated, we incur virtually total data loss and are left with mere estimates/CIs for the normal distribution, rather than the data itself.

The following instruction is specific to estimating uncertainty of differences in means.

In programming terms,

- Start by creating SE values for each group. Here, it may be a good exercise to see how differences in the number of data points translates to differences in how wide/narrow the SE ranges are.
- Then, generate simulations of the sample, usually in the form of a tibble
  - Define the number of simulations, n, to run
  - Then, retrieve n draws from the normal distribution $X \sim \mathcal{N}(\hat{\theta}, \mathrm{SE}^2)$

Bootstrapping

The bootstrapping approach to uncertainty estimation uses resampling *with replacement* to generate a sampling distribution of the estimates. This utilises and extends information provided in the sample to create a sampling distribution of the parameter, rather than relying on an exogenous assumption (e.g., normality) which may or may not be true.

The mechanics of resampling with replacement means that it works well with small-n and non-normal distributions, making it the best method of the three. Bootstrapping's only real deficit is its computational power requirements. With current computing standards, it is essentially no longer a problem. Thus, bootstrapping is the go-to method for every form of parametric/non-analytic inference.

Bootstrappping consists of the following steps:

- First, pull all the data points by the grouping variable/s of interest.
- Then, create the actual simulations of sampling by creating a tibble using `expand_grid()`. Here, the number of rows being defined by the number of bootstrap simulations to be conducted, and the number of columns is the sample size.
- Afterwards, group the *tibble* data by row (by each bootstrap simulation).
- Lastly, summarise the mean *including* the argument `replace = TRUE`.

Misc

- Sometimes, reflecting on the DGP is important in order to grasp 'oddities' in our data.
  - e.g., when we are confident that our DGP is random enough for a normal distribution but it doesn't look normal
  - It could well be the fact that the numbers were rounded and the range so narrow that the resulting data loss compressed the distribution into a janky shape (e.g., polls)
- When you write equation code, write it in mathematical form first
- Central limit theorem

- Our theoretical assumption is IID -> every sample drawn from distribution is drawn in the same way (every time we draw a sample, it has to happen in a way where we cannot tell whether we're drawing a sample for the first or tenth time)
  - For practical reasons this is violated in survey sampling (since it's pointless surveying someone again)
  - But fundamentally, this is what is going on in bootstrapping (sampling with replacement is a necessary consequence of i.i.d.)
- Interpreting CIs: e.g., I'm (x%) confident that the assumption holds, and the assumption is unlikely to be violated

Week 4

Basic intuition

Assume a two-dimensional scatterplot for which we assume a linear relationship between the two variables. The entire set of all the points, (X_i, Y_i), in the scatterplot of dependent variable Y against explanatory variable X has both a set of x-values, x_i, and y-values, y_i. These values therefore have a mean along both the x-axis, $\bar{X} = \bar{x}$, and the y-axis $\bar{Y} = \bar{y}$. Given that we expect most (if not all!) data points to be offset from $\bar{X}$ and $\bar{Y}$ (and therefore, their respective values offset from $\bar{x}$ and $\bar{y}$)[1], we expect to see a difference between the point values and the mean value, $y_i - \bar{y}$.

Some of this deviation is going to be explained by the line, which gives us the predicted values $\hat{x}, \hat{y} \quad \forall x \in X$, but social science data is (at least overwhelmingly) not perfectly linear. Therefore, we additionally expect residuals, ϵ_i, as the deviations from the mean that are unexplained by the line, i.e., $y_i - \hat{y}$. Given the linear regression formula

$$\hat{y} = \beta_0 + \beta_1 \hat{x}$$

we can re-formulate E_i as

$$\begin{aligned} \epsilon_i &= y_i - \hat{y} \\ &= y_i - (\beta_0 + \beta_1 \hat{x}) \end{aligned}$$

Going forward, it is important to remember that the sum of residuals equals zero. Why is the sum of residuals $\epsilon_i = \sum(y_i - \bar{y}) = 0$, as well as the sum of x-axis deviations from the mean $\sum(x_i - \bar{x}) = 0$?

Let m_i be a series of values $m_1, m_2, ..., m_n$. Let its mean be $\bar{m} = \frac{1}{n}\sum_{i=1}^{n}(m_i)$. Therefore, $m_i = \bar{m} + (m_i - \bar{m})$, the latter of which we can abstract away as $\delta_i = m_i - \bar{m}$. This means that

[1] This concerns an ultimately technical distinction, but given the importance of separating concepts from their values, I decided to go through the trouble of writing this out.

$$
\begin{aligned}
\bar{m} &= \frac{1}{n}\sum_{i=1}^{n}(\bar{m}+\delta_i) \\
&= \frac{1}{n}(\sum_{i=1}^{n}\bar{m}+\sum_{i=1}^{n}\delta_i) \\
n\bar{m} &= \sum_{i=1}^{n}\bar{m}+\sum_{i=1}^{n}\delta_i \\
&= n\bar{m}+\sum_{i=1}^{n}\delta_i \\
\sum_{i=1}^{n}\delta_i &= n\bar{m}-n\bar{m} \\
&= 0
\end{aligned}
$$

Let either $m_i = x_i$ and $\delta_i = x_i - \bar{x}$ or $m_i = y_i$ and $\delta_i = y_i - \bar{y}$. Therefore

$$\sum_{i=1}^{n}(x_i-\bar{x}) = \sum_{i=1}^{n}(y_i-\bar{y}) = 0 (4.1)$$

This finding motivates us to operationalise 'minimum residuals' differently than simply adding up their values. The standard method of doing this is by squaring them. We do this for three reasons:

1. We remove the cancelling effect of negative-sign residuals (common to techniques like LAV)
2. We penalise outliers more in calculating the line of best fit (specific to power transformation)
3. It is more mathematically tractable, allowing partial derivation to obtain coefficient values

Linear regression

The above minimisation problem can be formulated as

$$\min_{\hat{\beta}_0,\hat{\beta}_1}\sum_{i=1}^{n}(y_i-\hat{\beta}_0-\hat{\beta}_1 x_i)^2$$

Given that we have the entire term squared, we can (partially) differentiate to find out the first-order conditions given a change on *either* $\hat{\beta}_0$ or $\hat{\beta}_1$.

$$
\begin{aligned}
\frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 &= \sum_{i=1}^{n} 2(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(-1) \text{ by chain rule} \\
&= -2 \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) \qquad \stackrel{!}{=} 0 \\
\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) &= 0 \\
\sum_{i=1}^{n} y_i - n\hat{\beta}_0 - \sum_{i=1}^{n} \hat{\beta}_1 x_i &= 0 \\
\sum_{i=1}^{n} y_i - \hat{\beta}_1 \sum_{i=1}^{n} x_i &= n\hat{\beta}_0 \\
\hat{\beta}_0 &= \frac{1}{n} \sum_{i=1}^{n} y_i - \hat{\beta}_1 (\frac{1}{n} \sum_{i=1}^{n} x_i) \\
&= \bar{y} - \hat{\beta}_1 \bar{x} \qquad (4.2)
\end{aligned}
$$

Given this formulation of $\hat{\beta}_0$, any OLS estimator thus passes through both the x-axis and y-axis mean values. Now, to find $\hat{\beta}_1$, repeat the FOC differentiation but on $\hat{\beta}_1$

$$\frac{\partial}{\partial \hat{\beta}_1} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 = \sum_{i=1}^{n} 2(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(-x_i) \qquad \text{by chain rule}$$

$$= -2 \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(x_i) \qquad \stackrel{!}{=} 0$$

$$\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)(x_i) = 0$$

$$\sum_{i=1}^{n} (y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i)(x_i) = 0 \qquad \text{sub in (4.2)}$$

$$\sum_{i=1}^{n} (y_i - \bar{y} + \hat{\beta}_1 \bar{x} - \hat{\beta}_1 x_i)(x_i) = 0$$

$$\sum_{i=1}^{n} (y_i - \bar{y} - \hat{\beta}_1 x_i + \hat{\beta}_1 \bar{x})(x_i) = 0$$

$$\sum_{i=1}^{n} ((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}))(x_i) = 0$$

$$\sum_{i=1}^{n} (y_i - \bar{y})(x_i) - \hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i) = 0$$

$$\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i) = \sum_{i=1}^{n} (y_i - \bar{y})(x_i)$$

$$\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(\bar{x} + x_i - \bar{x}) = \sum_{i=1}^{n} (y_i - \bar{y})(\bar{x} + x_i - \bar{x})$$

$$\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(\bar{x}) + \hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x}) = \sum_{i=1}^{n} (y_i - \bar{y})(\bar{x}) + \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})$$

$$\bar{x}\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x}) + \hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x}) = \bar{x} \sum_{i=1}^{n} (y_i - \bar{y}) + \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})$$

$$\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x}) = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) \qquad \text{sub in (4.1)}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x})} \qquad (4.3)$$

$$= \frac{\text{Cov}(x, y)}{\text{Var}(x)}$$

$$\beta_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

The slope coefficient, $\hat{\beta}_1$ represents the *marginal coefficient*

Goodness of fit

Goodness of fit is how well the model captures variance in the data (i.e., how close the explained sum of squares/ESS gets to the total sum of squares/TSS).

TSS = ESS + RSS

ESS is $(\hat{Y} - \bar{Y})$ and RSS is $(Y_i - \hat{Y})$.

R^2 is a measure of how well the model at hand captures variation between the two variables of interest. It is operationalised as the ratio between the explained and total sum-squares (ESS divided by TSS),

$$R^2 = \frac{\hat{Y} - \bar{Y}}{Y_i - \bar{Y}}$$

Given that ESS = TSS - RSS, we can alternatively formulate this as 1 - (RSS/TSS), i.e.,

$$R^2 = 1 - \frac{Y_i - \hat{Y}}{Y_i - \bar{Y}}$$

Adjusted R^2

Formally, the adjusted R^2 is given as

$$\begin{aligned} \text{adj. } R^2 &= 1 - (1 - R^2)(\frac{n-1}{n-k-1}) \\ &= (1 - R^2)(\frac{n-1}{n-(k+1)}) \end{aligned}$$

Here, we see the following things happening:

- The sample size minus one is divided by the degrees of freedom, with $(k+1)$ representing the number of parameters (k IVs and the intercept)
- Given k is strictly positive in a regression context, $n - k - 1 < n - 1$
- This means that $\frac{n-1}{n-k-1} > 1$
- Multiplying $1 - R^2$ by $\frac{n-1}{n-k-1}$ therefore inflates its value, decreasing the value of the overall expression
- We therefore get a smaller R^2 value that decreases with more IVs, ceteris paribus

Assumptions

A1. Linearity

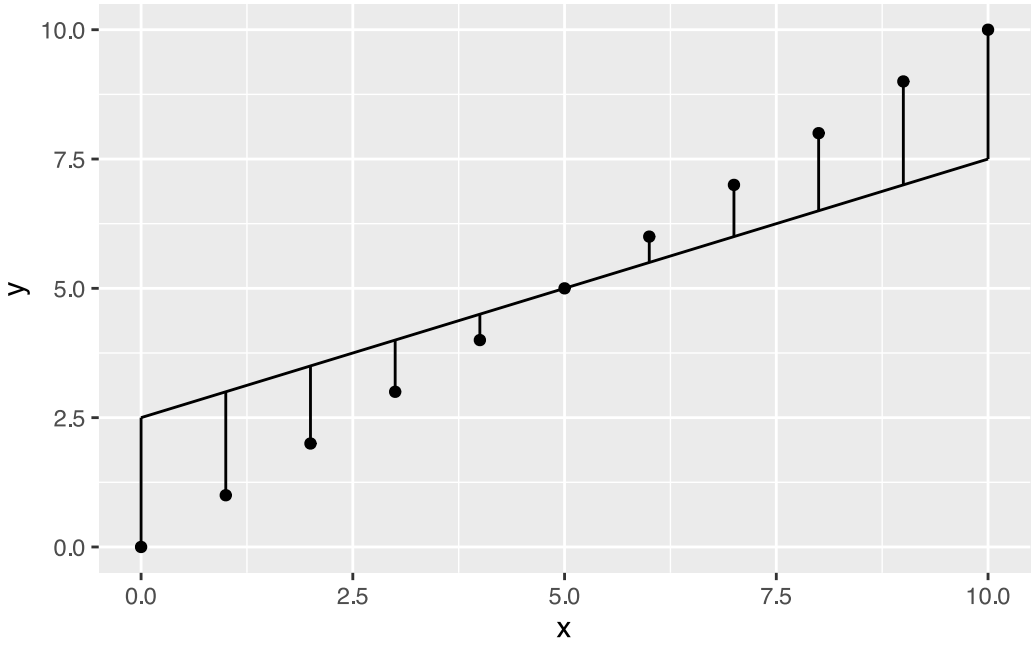
The coefficients should be linear. This is self-explanatory and violation of linearity implies, at minimum, violation of A2.

A2. Zero conditional mean

The expected value of the standard error of i, given some X, is 0:

$$(\epsilon_i | X) = 0$$

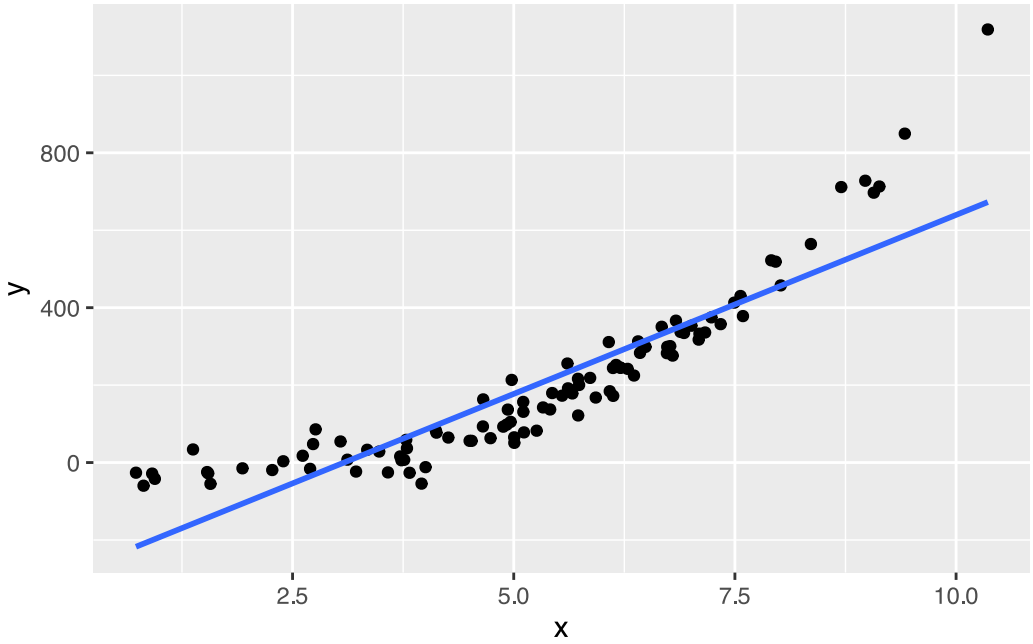
This implies that the error terms should not be correlated with the independent variable. We will cover two potential scenarios involving violation of this assumption. The first is when the slope is misidentified (and therefore the marginal effect biased):



Regressing the residuals in the graph above against Y gives us a (statistically significant) correlation, when we expect none or a non-significant one:

	(1)
Intercept	-2.500
X-value	0.500
Num.Obs.	11
R2	1.000

The second is when relationship is not linear (which also means A1 is violated):

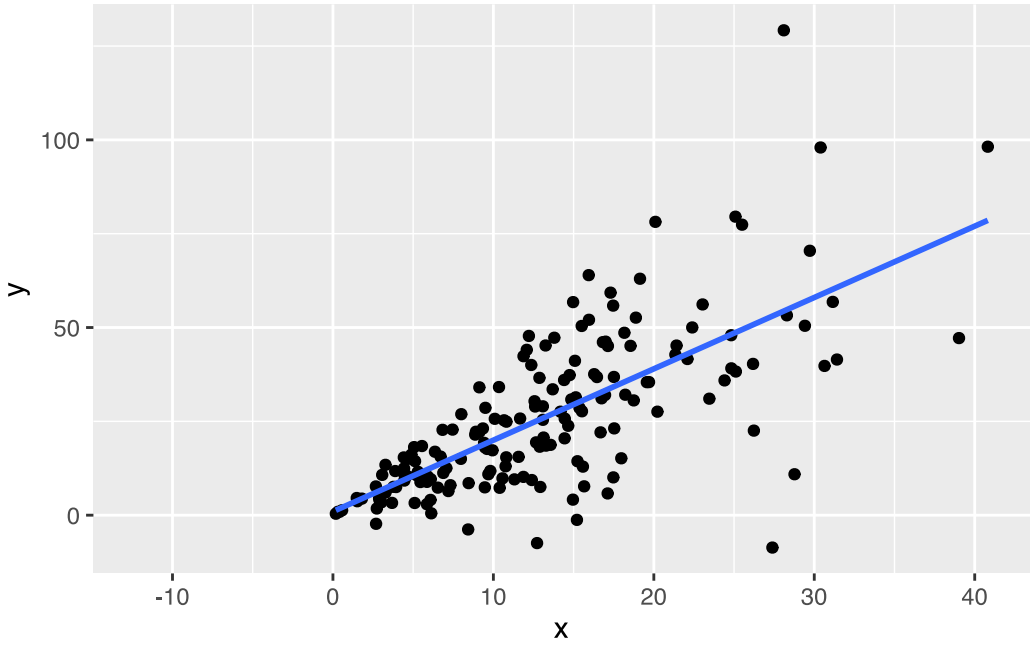


A3. Constant variance (homoscedasticity)

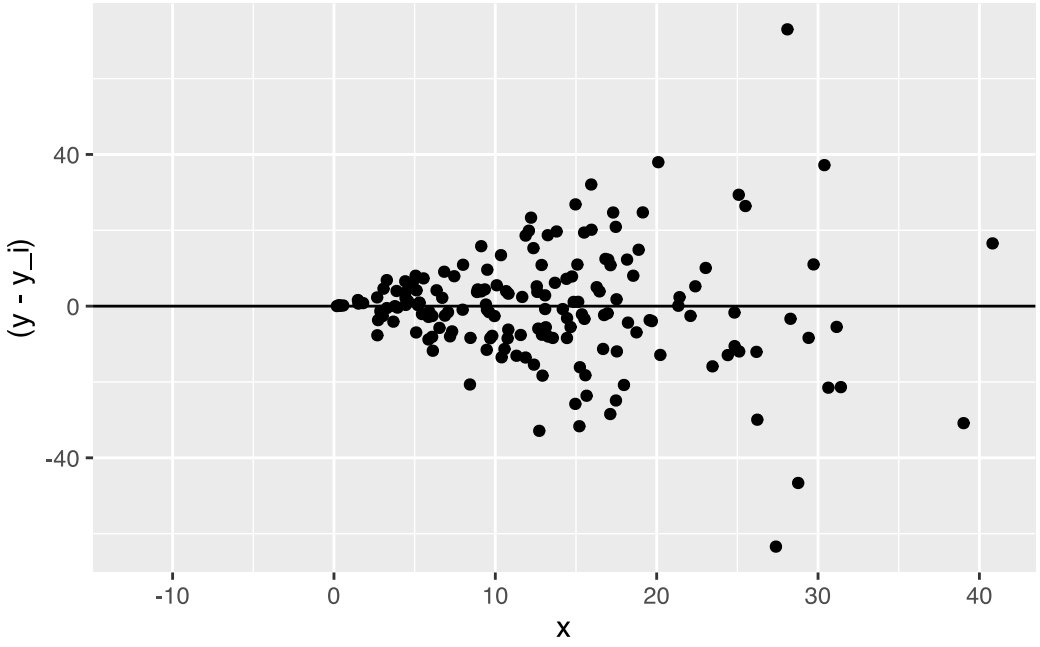
The variance should remain constant regardless of X, i.e.,

$$\text{Var}(\epsilon_i|X) = \sigma^2$$

As can be seen below:



A way to assess this is by creating a residual plot, which detects systematic variation not captured by the model. Such a plot graphs the error against the x-value.



A4. Uncorrelated errors

A violation of the uncorrelated errors assumption means that the estimated standard errors are underestimated, i.e., narrower than they should be. This may result in erroneously calling a relationship statistically significant when it isn't.

Broadly speaking, there are three types of violation here. The first involves positively correlated errors:

The second involves negatively correlated errors:

The third involves accidental

A5.

The error terms are distributed normally

Week 5

Multiple regression

When we presume independence between the IVs (i.e., no interaction), the β-coefficient of each IV is interpreted as the marginal effect of the associated IV on the DV *holding all else equal.*

The minimisation problem in a multiple regression involving k independent variables is a simple extension of the minimisation problem in bivariate regression, except with $k + 1$ equations to solve

To manually retrieve the controlled effect of an IV of interest on the DV, we conduct the following steps:

- Regress the confounder on the DV and retrieve the residual of this regression
- Regress the confounder on the IV and retrieve the residual of this regression
- Regress the residuals on one another; the slope coefficient is the marginal effect of the IV on the DV *ceteris paribus*

Directed Acyclic Graphs (DAGs)

Directed acyclic graphs (DAGs) are node-edge graphs that formalise our theory/hypothesis of some causal mechanism between an *exposure*, our IV of interest, and an *outcome*, our DV of interest. They are directed because the edges are unidirectional, often represented by arrows linking an *ancestor* to a *descendant*. They are 'acyclic' because none of the arrows emanating out of the exposure variable comes back to the exposure. The absence of an arrow between two variables implies an argument of no causality, which is as theoretically important as the presence of them.

Other than exposure or outcome variables, there are three other general classes of nodes

- Mediators, nodes on the causal pathway between the exposure and outcome; divided into
  - Partial mediators, mediators which do not account for the entire causal pathway
  - Full mediators, mediators which serve as intermediaries for the entire causal effect
- Confounders, nodes that are ancestors of the exposure and the outcome/a mediator and therefore induce a non-causal association between the two
- Colliders, nodes where two or more causal effects meet

Controlling for mediators and confounders blocks the path, whereas controlling for a collider opens it up.

Control variables

A control variable that decreases estimation bias must fulfil *all* of the following criteria:

- Is a confounder or a proxy of a(n unobserved) confounder
- Is not a collider

Aside from having a causal effect on the exposure, it can have a causal effect on either the outcome or some mediator.

A control variable increases estimation bias when it meets *any one* of the following criteria:

- Is a collider, wherever relative to the exposure/outcome it is (including as an ancestor)
- Is downstream of the causal effect, even if it's 'only' downstream of the mediator
- Is an ancestor of the exposure when a confounder remains unobserved/uncontrolled for (bias amplification)
- Is a mediator

Some controls do not change estimation biases but have other (positive/negative) effects, such as

- (+ve) A different IV that does not also cause variation on the exposure *or mediator*

- Increases estimate precision (lower variance) by reducing variation on the outcome *or mediator*
- (+ve) A variable mediating the relationship between the exposure and some collider W
  - Gives the W-specific outcome of the exposure on the outcome
- (-ve) An ancestor of the exposure that doesn't also cause variation on the outcome
  - Decreases estimate precision (higher variance) by reducing variation on the exposure
- (-ve) A different DV that is not causally connected to the outcome (of interest)
  - Decreases estimate precision (higher variance) by reducing variation on the exposure

Week 6

Confidence intervals

What are confidence intervals?

Confidence intervals (CIs) are an expression of uncertainty about some estimate's precision, constructed such that the true population mean will be between an analytically calculated (class of) interval(s) some high percentage of the time, usually 95%. The level of 'error' (that is, the probability that the population mean is outside of this class of ranges) is called the 'significance level', $\alpha \in (0, 1)$.

Now to the construction of the confidence interval itself: let some α and let δ be a (+)ve number. The CI around some marginal effect β_j is defined as

$$\Pr(\hat{\beta}_j - \delta \leq \beta_j \leq \hat{\beta}_j + \delta) = 1 - \alpha \quad (6.1)$$

That is, the probability that the true population mean is between (the estimate ± δ) equals one minus the significance level.

A remark on CI interpretations

Let $\hat{\beta}_j$ have some sign, whether it be $(+)$ or $(-)$. If the CI doesn't include the zero point, we accept some 'possibility of variation' (i.e., imprecision) in the *value* of the actual population parameter, but we are $(1 - \alpha)$ confidence *a priori* that the population parameter agrees with our *interval estimate*[2], and perhaps slightly more so in terms of the sign. Ergo, one can say that they qualitatively agree, although to some this might not be sufficient[3].

If a CI does include the zero, however, then it is a sufficient (albeit not *always* necessary[4]) condition for qualitative disagreement. The CI including the zero point means that we have an *a priori*

[2]To reiterate, this means that in the long run, $(1 - \alpha)$ of all *repeated trials* will contain the population interval given this CI *specification*.

[3]Whether this is a sufficient criterion for qualitative agreement, or whether the absolute values of the estimates themselves actually matter, is for sound judgment to decide. Even then, sound judgment should still be deployed to determine whether the possible values in the confidence intervals represent wide enough of a 'span' to change its associated substantive conclusions.

[4]See preceeding footnote

epectation that there is a non-negligible[5] probability of the coefficient/marginal effect being either

- null, or
- having the opposite effect to the one expected.

The two ways to calculating CIs

One way to calculate CIs is to presume known population variance (ergo, also the coefficient SEs) and assume a normal sampling distribution. We thus construct our CIs with $\delta \approx 1.96SE(\beta_j)$. This is also generally applicable whenever we have $n > 120 + k + 1$ observations ($n - k - 1 >$ 120 DOF), with k being the number of explanatory variables.

However, we don't always have large enough samples - or perhaps, we have 'too many explanatory variables'. We can relax this assumption of known population variance and instead use a different (non-normal) distribution, the t distribution. This is given as

$$t^* = \frac{\hat{\beta}_j - \beta_j}{\mathrm{SE}(\hat{\beta}_j)} \tag{6.2}$$

$$= \frac{\hat{\beta}_j - \beta_j}{\sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^{n}(x_{ji} - \bar{x})^2}}} \sim t_{(n-k-1)}$$

with the SE recalculated to utilise the estimated variance, causing the test statistic to no longer be normally distributed. The t distribution depends on the DOF in such a way that it is similar enough to the normal distribution with a high enough DOF (usually, more observations) for the normal distribution to be usable. We now have the CI

$$\Pr(t_{(\frac{\alpha}{2})} \leq t^* \leq t_{(1-\frac{\alpha}{2})}) = 1 - \alpha \tag{6.3}$$

Recall equation (6.1). In words, we understand this as $1 - \alpha$ being the probability that the true population parameter is between estimate $- \delta$ and estimate $+ \delta$. Therefore, the boundary values of the CI signify $\beta_j = \hat{\beta}_j - \delta$ and $\beta_j = \hat{\beta}_j + \delta$ respectively. Therefore, the δ value at the lower bound is

$$\delta = \hat{\beta}_j - \beta_{jl}$$

and the δ at the upper bound is

$$\begin{aligned} \delta &= \beta_{ju} - \hat{\beta}_j \\ &= -(\hat{\beta}_j - \beta_{ju}) \end{aligned}$$

and therefore

[5] Whether some level of significance, often 5%, constitutes a negligible percentage is a well-worn debate topic, though I certainly don't think it can ever be 'too worn'

$$\hat{\beta}_j - \beta_{jl} = -(\hat{\beta}_j - \beta_{ju}) \quad (6.4)$$
$$(= \delta \quad)$$

Revisiting equation (6.3), we verbally understand the equation as the probability that t^* is larger than or equivalent to the t-value for some DOF at $\frac{\alpha}{2}$[6] and smaller than or equal to t at $(1 - \frac{\alpha}{2})$.

Revisiting equation (6.2) and bringing (6.4) gives us $\delta = t^* \ \mathrm{SE}(\hat{\beta}_j)$, where the lower boundary is

$$\beta_{jl} = \hat{\beta}_j + t_{(\frac{\alpha}{2})} \ \mathrm{SE}(\hat{\beta}_j)$$

and the upper boundary

$$\beta_{ju} = \hat{\beta}_j + t_{(1-\frac{\alpha}{2})} \ \mathrm{SE}(\hat{\beta}_j)$$

Given that $\beta_{ju} = \hat{\beta}_j + \delta = \hat{\beta}_j + t^* \ \mathrm{SE}(\hat{\beta}_j)$ corresponds to the upper boundary, we can infer that $t^* = t_{(1-\frac{\alpha}{2})}$. By symmetry,

$$t_{(\frac{\alpha}{2})} \ \mathrm{SE}(\hat{\beta}_j) = -\delta$$
$$\hat{\beta}_j + t_{(1-\frac{\alpha}{2})} \ \mathrm{SE}(\hat{\beta}_j) = \hat{\beta}_j - t_{(\frac{\alpha}{2})} \ \mathrm{SE}(\hat{\beta}_j)$$

and therefore

$$\Pr(\hat{\beta}_j - t_{(1-\frac{\alpha}{2})} \ \mathrm{SE}(\hat{\beta}_j) \leq \beta_j \leq \hat{\beta}_j + t_{(1-\frac{\alpha}{2})} \ \mathrm{SE}(\hat{\beta}_j)) = 1 - \alpha \quad (6.5)$$

Errors, errors

Statistical inference essentially boils down to deciding between two alternative (substantive) interpretations: either there is no ('null') effect, H_0, or there is an effect, H_A. The latter is the 'alternative' hypothesis, hence the A subscript.

With a 95% CI (I hope what this means is clear enough to the reader by now), H_0 being true implies that we falsely reject H_0 ('false positive') 5% of the time *in the long run*. That is, $\Pr(\text{reject } H_0 | H_0) = 0.05$.

The α-bsolutist way...

In testing the statistical significance of a regression coefficient, we want to determine whether our estimated coefficient is non-null/non-zero, i.e., $\hat{\beta}_j \neq 0$. Therefore, we use a two-tailed hypothesis to test against H_0:

- H_0: $\beta_j^* = 0$, where $*$ denotes 'test'[7]
- H_A: $\beta_j^* \neq 0$

Thus, we construct t^* given our hypotheses about $\hat{\beta}_j$. Bringing back equation (6.2) and given a baseline expectation of null (i.e., $\beta_j^* = 0$),

[6] two-tailed, therefore the probability of the true population param lying outside the CI, which is what α essentially is, is divided equally for each of the two tails

[7] See t^* in the previous section

$$t^* = \frac{\hat{\beta}_j - \beta_j^*}{\text{SE}(\hat{\beta}_j)}$$
$$= \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)} \sim t_{(n-k-1)}$$

We reject H_0 at the α level if $|t^*| > t_{(1-\frac{\alpha}{2}[,n-k-1])}$.

...and the p-retty nuanced way

So far, we've framed statistical significance in binary terms, i.e., whether our test statistic clears some 'arbitrary' hurdle. What if different people and different venues set different hurdles? The concept of p-values reframes the expression of uncertainty and states it as the 'maximum height of the hurdle' which our test statistic can clear.

To return to our α example, our hurdle is $t^* = t_{(1-\frac{\alpha}{2},n-k-1)}$. When it is cleared, we are $100(1-\alpha)$ confident of a CI. So, what about the reverse? The probability that we misdiagnosed the situation and $|t^*|$ is actually smaller than $|t_{(n-k-1)}|$[8]?

Once again, $\Pr(t^* \geq t_{(1-\frac{\alpha}{2},n-k-1)}) = 1-\alpha$. Generalised,

$$\Pr(t_{(n-k-1)} \leq t^*) = 1-p$$

The p-value therefore takes the exact opposite scenario; neither $>$ nor $=$, but just $>$:

$$\Pr(t_{(n-k-1)} > t^*) = p$$

$\Pr(t > t^*)$ denotes the area to the right of t^* given $n-k-1$ DOF. By symmetry,

$$p = \Pr(|t_{(n-k-1)}| > |t^*|)$$
$$= 2\Pr(t_{(n-k-1)} > |t^*|)$$

The above gives us the smallest significance level at which H_0 can be rejected, given the resulting $t^*_{(n-k-1)}$.

Note: the p-value is the likelihood of seeing a t-score at least as extreme as the test statistic given a null world, which is in turn equivalent to the probability of rejecting H_0 given H_0 is true.

Categorical/binary variables

'Qualitative' (nominal, sometimes ordinal) explanatory variables[9] can be reduced to one, or a series of, binary (often called 'dummy') variables. When reducing/recoding to binary variables, it is good practice to

- Code the two categories within each binary variable as 0 and 1,
- Assign the 0 value to some baseline, and
- Name variables after the non-baseline ('1') category.

[8] Since we're not dealing with αs here, $\frac{\alpha}{2}$ is irrelevant

[9] Regressing them as DVs is the topic of a future section

Evading the binary variable trap

Let's take a polytomous variable, such as 'party an individual voted for', as an example. If each category gets its own binary variable, the resulting binary variables will be perfectly collinear as they all take the value 1. Given k categories, we avoid this collinearity trap by splitting this into $k-1$ binary variables, with the baseline category taking on the 0 value in all binary variables. Note that coefficient interpretation is always relative to the baseline category.

For polytomous nominal variables, simply assign one binary variable for non-baseline category with a 1 for the *respective* category it identifies and zero for all the others. For ordinal variables, it may help to keep the 1 instead of turning it to a 0 as we go up the scale. This allows us to calculate the coefficients between levels with standard error calculation included.

Interpreting interaction coefficients

Let's use the following example:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 D + \beta_3 X_1 D + \epsilon$$

$$\frac{\partial Y}{\partial X_1} = \beta_1 + \beta_3 D$$

$$\frac{\partial Y}{\partial D} = \beta_2 + \beta_3 X_1$$

Focusing on X_1,

- When $D = 0$, the marginal effect of X_1 is β_1.
- When $D = 1$, the marginal effect of X_1 is $\beta_1 + \beta_3$.

Focusing on the intercept,

- When $D = 0$, the intercept is $beta_0$. Quite straightforward so far, however...
- When $D = 1$, $\beta_2 D = \beta_2$ is also a constant, and the intercept is therefore $beta_0 + \beta_2$

We notice two things:

- The marginal effect of X_1 changes in value with different values of D
- The marginal effect of D changes the value of the intercept

This constitutes the ground upon which we explore the rules to follow in interpreting interaction coefficients, as laid out by Thomas Brambor et al. (2007).

Include interaction terms

It should be clear why: doing otherwise obscures the interaction between both variables. This This is a rather trivial point, but it bears repeating because it is perhaps fundamentally connected to the error in interpretation often seen in political science literature which came out more than two decades ago (two subheadings down).

Include all constituent terms

A common mistake in conducting interaction regressions is to exclude e.g., the binary constituent of an interaction effect. Specifically in this context, not specifying all constituent terms forces the regression lines to share an intercept, yielding biased slope coefficients.

Do not interpret marginal coefficients as unconditional

A common flawed practice in social science is to simply report one IV's marginal effects as its own (independent) beta-coefficient. In what follows, I mathematically elaborate why this is incorrect. Suppose an equation with two independent variables, X and Z, which do not interact:

$$\begin{aligned} Y &= \omega_0 + \omega_1 X + \omega_2 Z + \mu \\ \frac{\partial Y}{\partial X} &= \omega_1 \end{aligned}$$

When X and Z do not interact, reporting beta coefficients independently is not a problem. Now let's assume they do and add an interaction term:

$$\begin{aligned} Y &= \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + \epsilon \\ \frac{\partial Y}{\partial X} &= \beta_1 + \beta_3 Z \\ \frac{\partial Y}{\partial Z} &= \beta_2 + \beta_3 X \end{aligned}$$

This highlights that the marginal effect of either IV is not its beta-coefficient alone.

Calculate substantively meaningful marginal effects and SEs

Oftentimes you'll only be dealing with a binary interaction variable. Assuming such a case involving two interaction variables, and given that you're calculating marginal effects primarily from the perspective of the continuous variable, it's a straightforward task to interpret the marginal coefficients:

$$\begin{aligned} & Y = \beta_0 + \beta_1 X_1 + \beta_2 D + \beta_3 X_1 D + \epsilon \\ (D = 0) \quad & Y = \beta_0 + \beta_1 X_1 + \epsilon \\ (D = 1) \quad & Y = (\beta_0 + \beta_2) + (\beta_1 + \beta_3) X_1 + \epsilon \end{aligned}$$

Theoretically, there are only two sets of values (and four values total, the intercepts and slopes when $D = 0$ and $D = 1$, respectively) you'd need to report.

Interpreting continuous interactions is trickier, but there are simple tricks (e.g., using extreme values of the interaction variable or plotting a graph of how the coefficient changes with the interaction variable). For a good example of this, see Golder (2005).

Additionally, interaction estimates should be accompanied by the correct standard error calculation formula. The standard error of X_1's marginal effect on Y, $\frac{\partial Y}{\partial X_1}$ is

$$\hat{\sigma}_{\frac{\partial Y}{\partial X_1}} = \sqrt{\mathrm{Var}(\hat{\beta}_1) + X_2^2 Var(\hat{\beta}_3) + 2X_2 \ \mathrm{Cov}(\hat{\beta}_1, \hat{\beta}_3)}$$

Week 7

Hi

Week 8

Revisiting OLS assumptions

1. Linearity; that is, reducible to the form $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_k x_k + \epsilon$
2. Zero conditional mean; that is, $E[\epsilon | x_1, x_2, ..., x_n] = 0$
3. Homoscedasticity; that is, $\text{Var}(\epsilon | x_1, x_2, ..., x_n) = \sigma_\epsilon^2$
4. Uncorrelated errors; that is, $\text{Cov}(\epsilon_i, \epsilon_j | x_i, x_j) = 0 \forall i \neq j$
5. X is non-stochastic
   - Every time some $X_i = x_i$ gets picked up in a sample, the value always comes out the same
   - Implication: there is no measurement error
6. Normally distributed errors; that is, $\epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2)$ and therefore $Y_i \sim \mathcal{N}(\mu_i, \sigma_\epsilon^2)$
   - Implies that the error terms and Y_i are spread along the exact same distribution
7. No multicollinearity between IVs (also known as 'covariates')

More properties

- The OLS estimator is unbiased, i.e., $E[\hat{\beta}_j] = \beta_j$ if
  - The errors have zero expectation, $E(\epsilon) = 0$
- Under full assumptions, the OLS has the minimum variance $\therefore$ the highest efficiency (BLUE)

Issue 1: Misspecification

Motivation: we want to find out the bias of our model/estimator, but we cannot when using real-world data because we don't know whether the model is misspecified or otherwise.

Implication: We assess OLS properties under the prior assumption of misspecification. We do this by simulating data.

Let the true parameters be known and the DGP be defined by the following equation:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$
$$\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$$

with

$$x_1 = \alpha x_2 + \nu$$
$$\nu \sim \mathcal{N}(0, \sigma_\nu^2)$$

For $\alpha \neq 0$, as $\sigma_\nu^2 \to 0$, we'll approach exact linear dependency.

Our goal here is to see how a misspecified model fares against the correct model given some perfectly knowable DGP. We want to know the size/direction of the bias in some coefficient, say β_1.

Given some misspecified model $Y = \beta_0^* + \beta_1^* x_1 + \epsilon$, we can show that

$$
\begin{aligned}
E[\beta_1^*] &= \beta_1 + \beta_2\hat{\delta} \\
E[\beta_1^*] - \beta_1 &= \beta_2\hat{\delta} && \equiv \text{OVB} \\
&= 0 && \Leftrightarrow (\beta_2 = 0) \vee (\hat{\delta} = 0)
\end{aligned}
$$

Monte Carlo simulation

- Computational approach to study what happens when any of the OLS assumptions are not met
- Estimates unknown quantities by repeatedly drawing random samples from a known or assumed population, usually by generating fresh draws from some pre-specified DGP
  - In this case, a distribution with fully known and pre-specified parameters
- It must have a
  - Systematic component (relationship between X and Y)
  - Stochastic component (random noise, usually $\mathcal{N}(\mu_i, \sigma_\epsilon^2)$)
- Key processes in a Monte Carlo study
  - Simulate data for the explanatory variable X
  - Use the assumed DGP, including ϵ, to generate the dependent variable Y
  - Estimate the model repeatedly and examine the resulting sampling distribution

Steps

1. Generate 'fake' data X (i.e., simulate it) and assume we know the
   - True coeffs β
   - True DGP, including the errors ϵ
2. Given X, **true** β, and ϵ as an `rnorm`, calculate Y from the assumed true linear model
3. Given Y and X, run an OLS and extract the **estimated coefficients** $\hat{\beta}_j$
4. Repeat steps 2. and 3. e.g., 1000 times and retrieve the empirical sampling distribution
   - Similar to QoI simulation steps, except simulation is now based on assumed truth to see what happens when assumptions are violated

Steps (specific to the previous example)

1. Set `nobs`
2. Set values for x_2 and generate values for x_1
   - Remember, x_1 is the sum of $\alpha x_2 + \epsilon \sim \mathcal{N}(0, \sigma_\nu^2)$
   - In programming terms, $x_2 =$ `rnorm(nobs, 0,` σ_ν^2`)`
3. Generate Y as $\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon$ *and* $\beta_0^* + \beta_1^* X_1 + \epsilon$
4. Draw a sample from this distribution, then run an OLS; retrieve $\hat{\beta}_1$ and $\hat{\beta}_1^j$
5. Run 3. and 4. until you get `nobs` samples and `nobs` sets of $\hat{\beta}_1$ and $\hat{\beta}_1^j$
6. Plot and compare the simulated sampling distributions for both

Issue 2: measurement error

Motivation: recall the previous setup of X. It was generated once and held constant in every simulation. There is no 'randomness' or 'error' in the X itself. What if that wasn't the case and the measurements were erroneous?

Implication: simulate errors as a robustness check and see how 'far' the error needs to get before H_0 can no longer be rejected at our significance level of interest.

- Let a model $Y = \beta_0 + \beta_1 X + u$ have an imperfectly observed $X^* = X + e$
- Since $X = X^* - e$, $Y = \beta_0 + \beta_1 X^* + (u - \beta_1 e)$
- Given the assumption of zero correlation between e and X, there must therefore be a non-zero correlation between e and X^*
- Given assumptions 4 and 5, the correlation between e and X^* will yield biased estimates
  - When estimating via OLS, we get the asymptotic property $\hat{\beta}_1 \rightarrow \frac{\sigma_u^2}{\sigma_u^2 + \sigma_e^2}\beta_1$
- This attenuation bias, introduced through measurement error, biases estimates towards 0
- In multiple regression, it may go both ways (attenuation or amplification)

Monte Carlo study for measurement error

- Simulate X without and with the measurement error, respectively
- With an asymptotic attenuation bias, the size of the measurement error can be defined by varying σ_e^2
- Expectation: sample measurement for _1 (for X^*) biased towards 0
- Robustness: which level of measurement error renders our coefficients insignificant

What if the measurement error affects our DV instead?

- Let $Y^* = Y + u$
- Then $Y + u = \beta_0 + \beta_1 X + e$ and $\therefore Y = \beta_0 + \beta_1 X + (e - u)$
- The coefficient β_1 remains unchanged, although the variance increases (less precise estimate)

Issue 3: Multicollinearity

- High correlation between IVs/covariates, not limited to pairwise
- Leads to 'instability' in regression estimates (effect is similar to having a small n)
- Recall sampling variance of the OLS estimator

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma_\epsilon^2}{\Sigma(x_{ij} - \bar{x}_J)^2(1 - R_j^2)}$$

  with
  - R_j^2 being the linear relationship between IVs, i.e., R^2 from a regression of x_j on all other IVs
  - σ_ϵ^2 being the error variance (noise)
  - $\Sigma(x_{ij} - \bar{x}_J)^2$ being thesample variance in x_j
- Given the above $1 - R_j^2$
  - As $R_j^2 \rightarrow 1$, $1 - R_j^2$ decreases, the denominator decreases, and therefore variance increases
  - Implication: more sample variation in xj can be explained by the other covariates
- Consequently, we get strongly correlated covariates, which can be interpreted as
  - There being little variance in x_i when x_{-i} are fixed, and therefore
  - There being little information about the impact of x_i on Y holding x_{-i} constant
- Strong, but less than perfect, multicollinearity increases the sampling variance of OLS coefficients

  - Increasing the SEs and the size of the CIs
  - Affecting hypothesis testing
- Nonetheless, this does not bias the coefficient

How to detect multicollinearity

- Correlation tables work for a first-pass indication, but pairwise correlations cannot reveal highly collinear linear combinations
- For diagnostics (and only for diagnostics), use $\text{VIF}_j = \frac{1}{(1-R_j^2)}$, the variance-inflation factor
  - Ultimately, thresholds are arbitrary and you can't improve the model using this information
- You can also create indices merging some of the IVs using factor analysis

Issue 4: Heteroscedasticity

Causes:

- Can be because of model misspecification
  - Transform IVs and potentially DV through log, square, square-root
  - Plot the residual against the respective variables to find candidates for transformation
- Can be due to outliers/high-leverage points (influential observations)

Issue 5: Influential observations

- Need to
  - Identify observations dominating the estimates
  - Assess how results change when such dominating observations are dropped

Terms

- Outlier: observations with *large residuals*
- Leverage: How far a covariate deviates from the mean
  - A high-leverage observation has an extreme value on the IV
  - It has an effect on $\beta_i, i \neq 0$ (the regression coefficients)
- Influence: TWE removing an observation changes the estimate of a coefficient

Cook's distance

Cook's distance is a measure of influence, aggregating both concepts of outliers and leverage

$$D_j = \frac{\Sigma_{j=1}^{n}(\hat{y}_j - \hat{y}_{j\setminus i})^2}{(k+1)\hat{\sigma}^2}$$

with - k being the number of covariates/IVs - $\hat{\sigma}^2$ being the estimated error variance - $\hat{y}_{j\setminus i}$ being the predicted value for j without observation i It is a measure of distance based on the *difference in predicted values with or without a specific observation.*

Note: it is model-specific (you cannot compare the Cook's distance values between different models)

Rules of thumb

- Fixed, often more relaxed: $D_j > 1$

- n-dependent, often stricter: $D_j > \frac{4}{n}$, with n being the sample size (Bollen & Jackman, 1990)

How to handle influential observations

- Create a binary variable in your dataset, which takes the value 1 when Cook's distance is 'too large' given the above guidelines
- Respecify the model to account for outliers. Do they have anything in common?
- Potentially, drop observations from analysis (and acknowledge this in your writeup). Do you still have the same results?